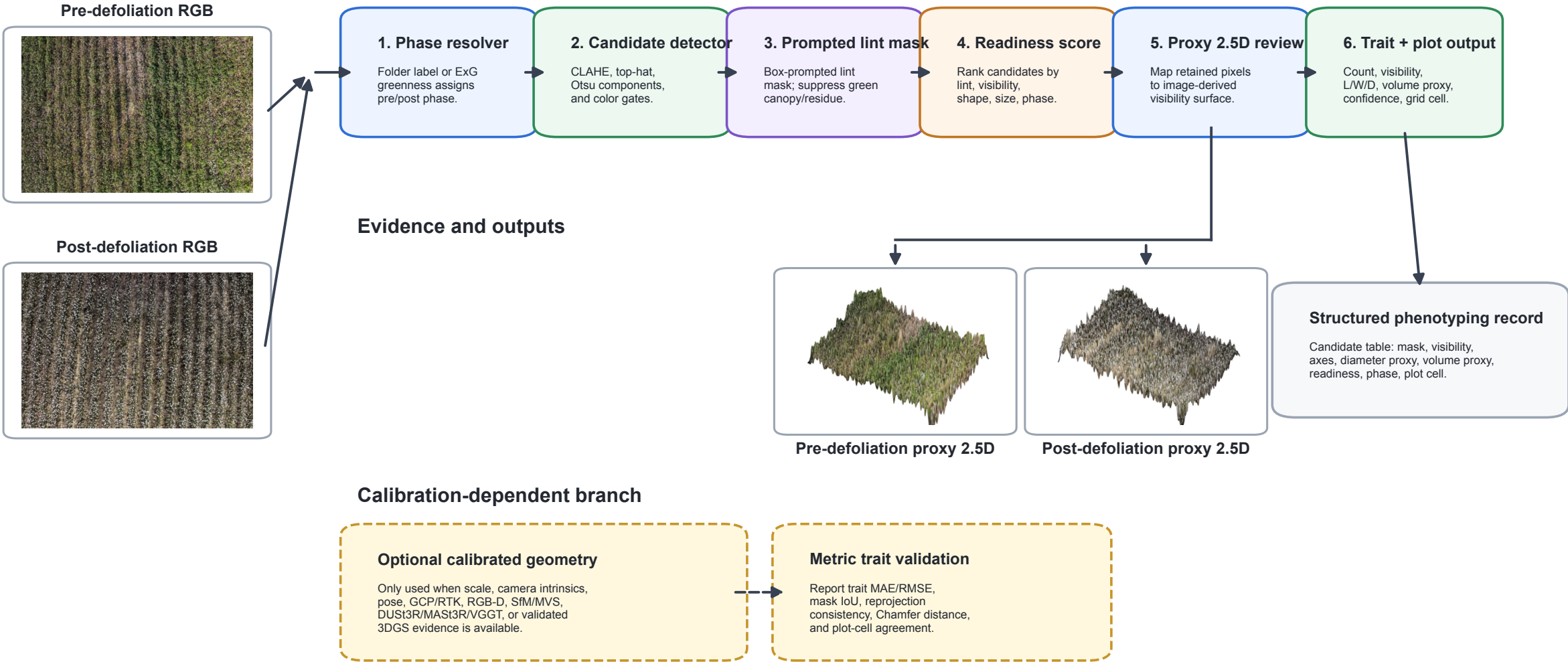


Mask-Guided Cotton Boll Phenotyping Architecture

Pre/post-defoliation UAV imagery → detector-prompted lint masks → morphology 2.5D review → conservative trait summaries

Inputs: aerial visibility intervention Implemented proxy pipeline



Solid arrows: implemented MVP/proxy analysis. Dashed arrows: calibration-dependent validation path.
Scientific boundary: 2.5D maps are image-derived aerial visibility surfaces; metric boll geometry requires calibrated scale/camera validation.